

# TEMPEST: Porewater SO<sub>4</sub>/Cl

202606 302-432 Samples

2026-09-22

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##Add Required Packages

## 0.1 Run Information

```
##### Run information - PLEASE CHANGE
Sample_Year = "2026"
Date_Run = "2026-09-25" #Date that instrument was run
Run_by = "Zoe Read" #Instrument user
Script_run_by = "Zoe Read" #Code user
run_notes = "Run looks good other than one bad chloride dup, likely due to a small chloride value.
TMP_FW_PW_C6_15cm_S04_20260615_Post had an unusually high chloride value for freshwater" #any notes f

##Sample data that was entered incorrectly
# The Old ID is the original, incorrectly-entered ID and the New ID is the correct ID to change it to.
Old_ID_1 = NA
New_ID_1 = NA

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name_cl = "Raw Data/COMPASS_TEMPEST_202606_302-432_Cl.txt"
raw_file_name_so4 = "Raw Data/COMPASS_TEMPEST_202606_302-432_S04.txt"

#Sample ID Path
SampleID_path = "Raw Data/TEMPEST_S04_20260501-20260612_corrected.csv"

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_TEMPEST_Processed_Cl_S04_202606_302-432.csv"

##### Log Files - PLEASE CHECK

#qaqc log file path for this year copied over from COMPASS GitHub
Log_path = "Raw Data/COMPASS_Synoptic_Cl_S04_QAQClog_2026.csv"

cat(run_notes)
```

```
## Run looks good other than one bad chloride dup, likely due to a small chloride value.
## TMP_FW_PW_C6_15cm_S04_20260615_Post had an unusually high chloride value for freshwater
```

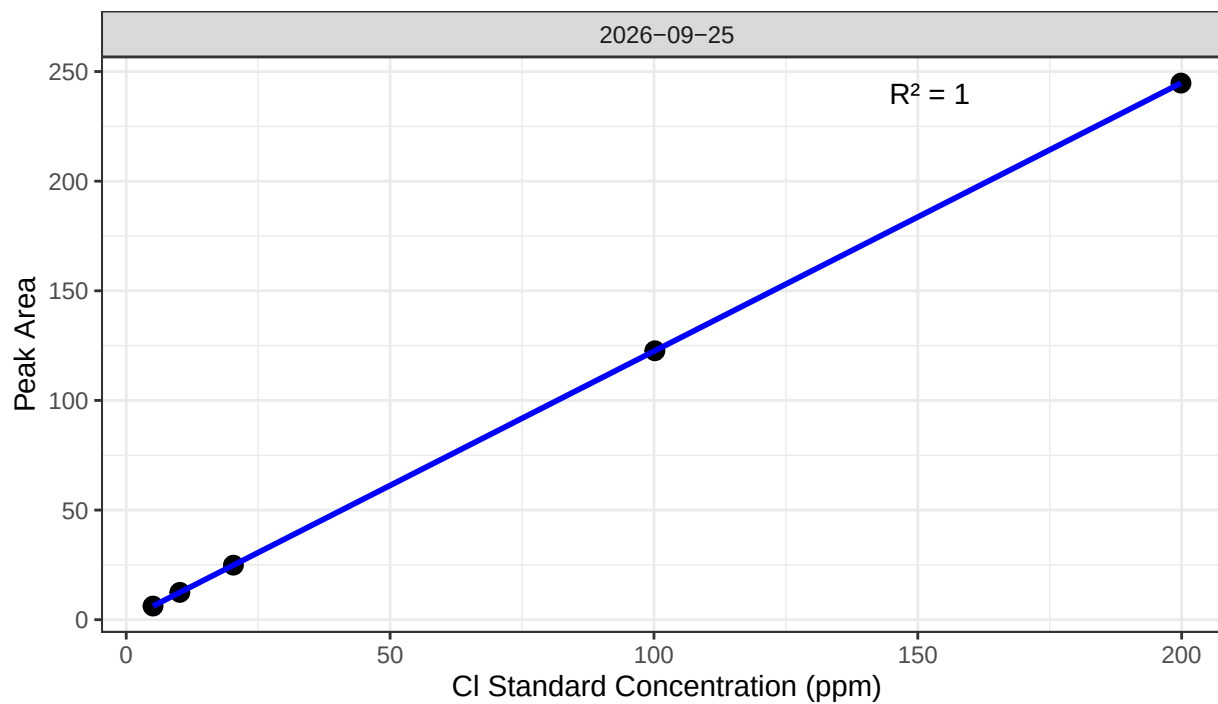
##Set Up Code - constants and QAQC cutoffs

##Import Sample Data

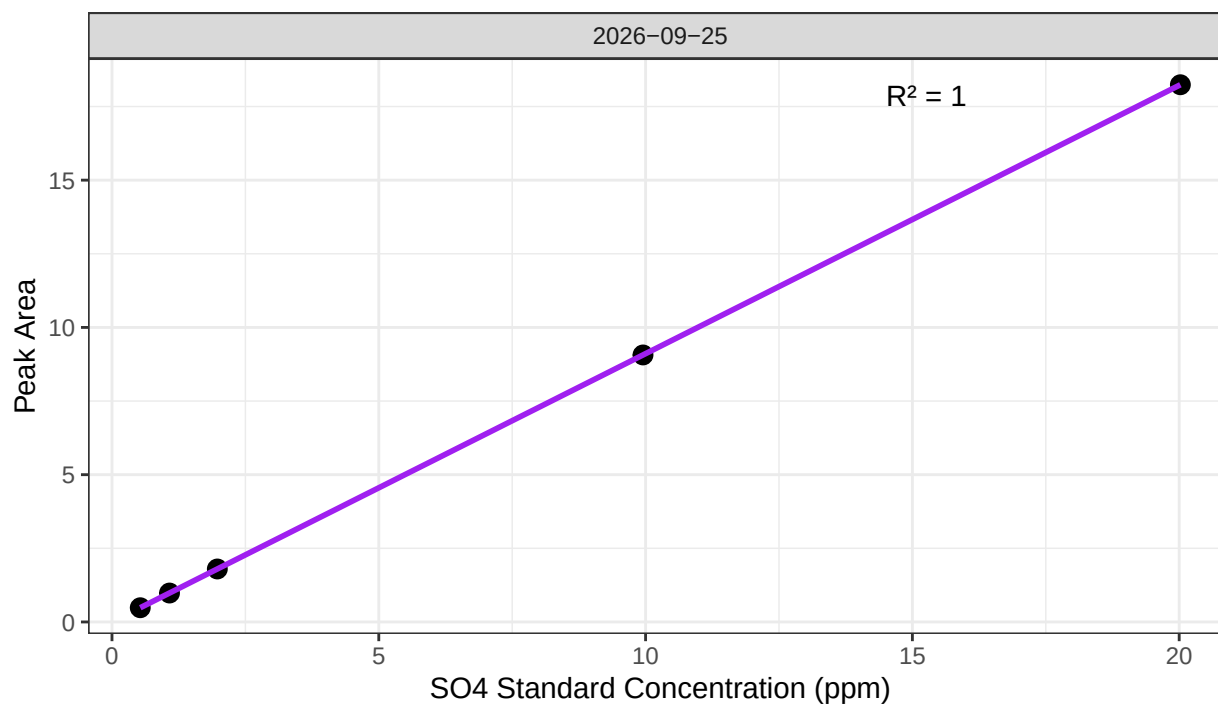
##Fix Sample IDs entered wrong

## 0.2 Assess Standard Curves

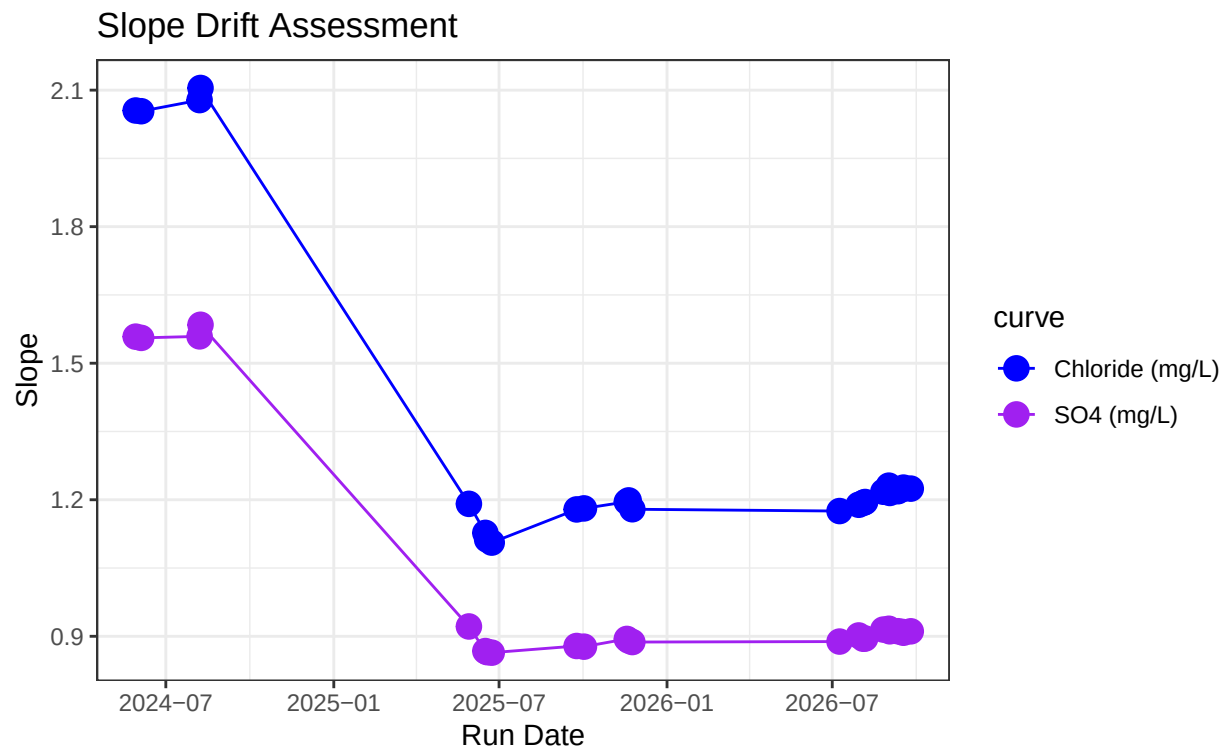
Chloride Std Curve



Sulfate Std Curve



```
## [1] "QAQC log file exists and has been read into the code."
```



```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

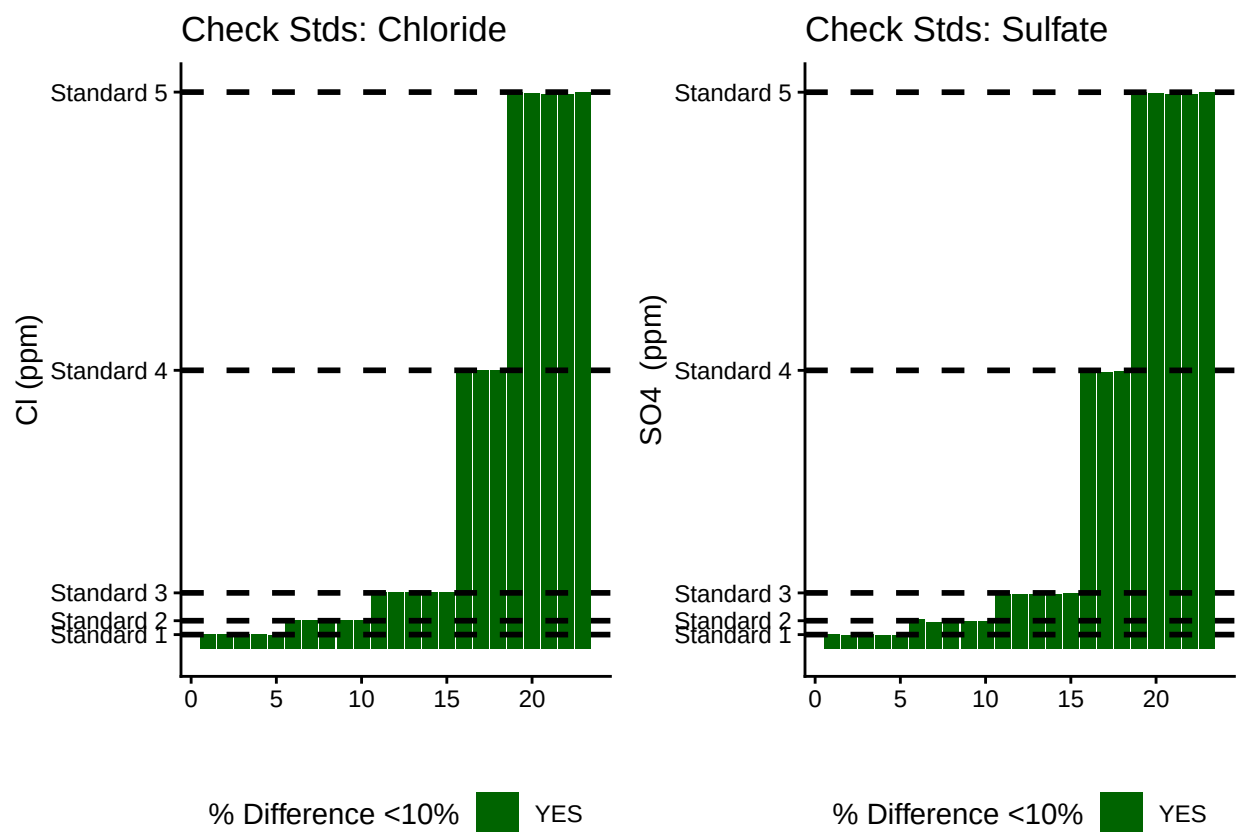
### 0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID mean_Cl sd_Cl cv_Cl flag_Cl
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1    5.07 0.0361 0.00712 Chloride Check Standard RSD within Range --
## 2 Standard 2   10.1 0.0373 0.00369 Chloride Check Standard RSD within Range --
## 3 Standard 3   20.3 0.0212 0.00104 Chloride Check Standard RSD within Range --
## 4 Standard 4  100. 0.0941 0.000941 Chloride Check Standard RSD within Range --
## 5 Standard 5  200. 0.335 0.00168 Chloride Check Standard RSD within Range --
```

```
## # A tibble: 5 x 5
##   sample_ID mean_SO4 sd_SO4 cv_SO4 flag_SO4
##   <chr>      <dbl> <dbl> <dbl> <chr>
## 1 Standard 1    0.500 0.0230 0.0460 Sulfate Check Standard RSD within Range --
## 2 Standard 2    0.996 0.0467 0.0469 Sulfate Check Standard RSD within Range --
## 3 Standard 3    1.97 0.00481 0.00243 Sulfate Check Standard RSD within Range --
## 4 Standard 4    9.95 0.0198 0.00199 Sulfate Check Standard RSD within Range --
## 5 Standard 5   20.0 0.0416 0.00208 Sulfate Check Standard RSD within Range --
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



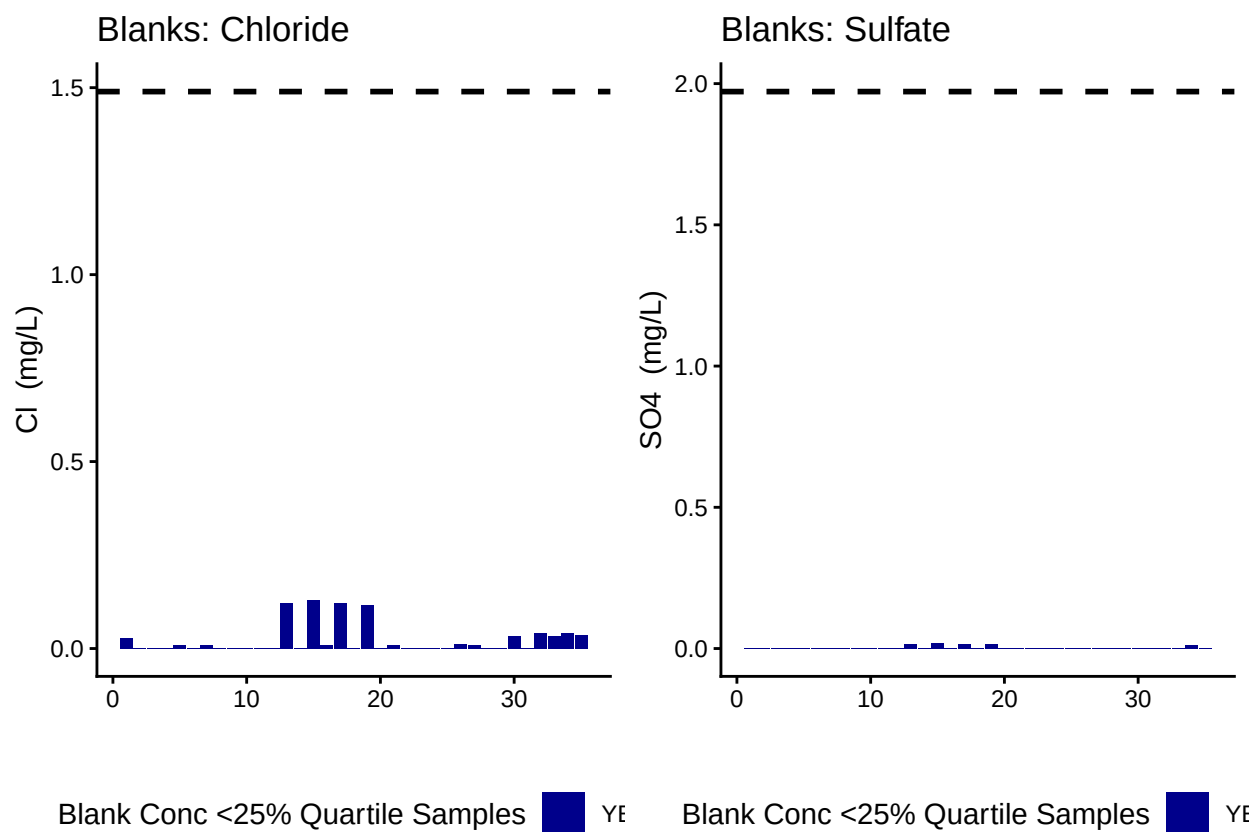
```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

#### 0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



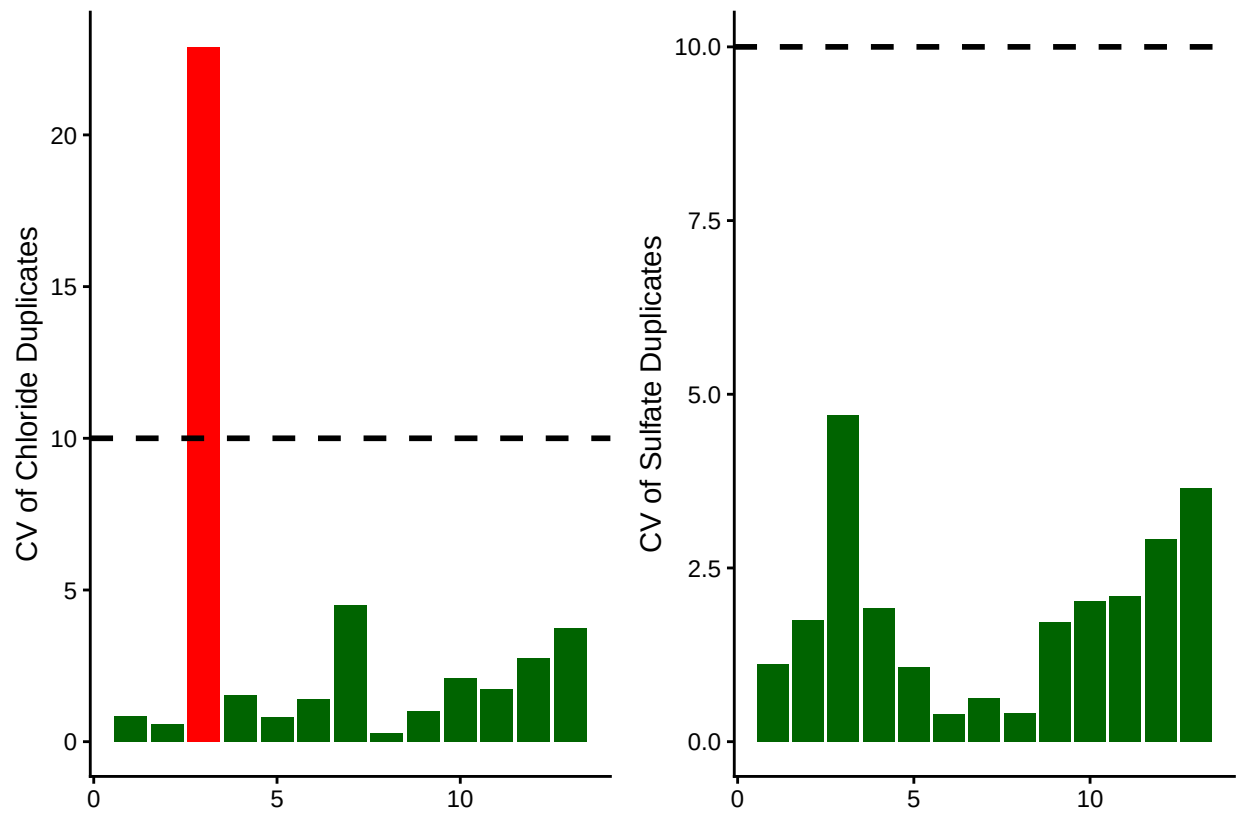
```
## Chloride blanks mean ppm:
```

```
## [1] 0.02182286
```

```
## Sulfate blanks mean ppm:
```

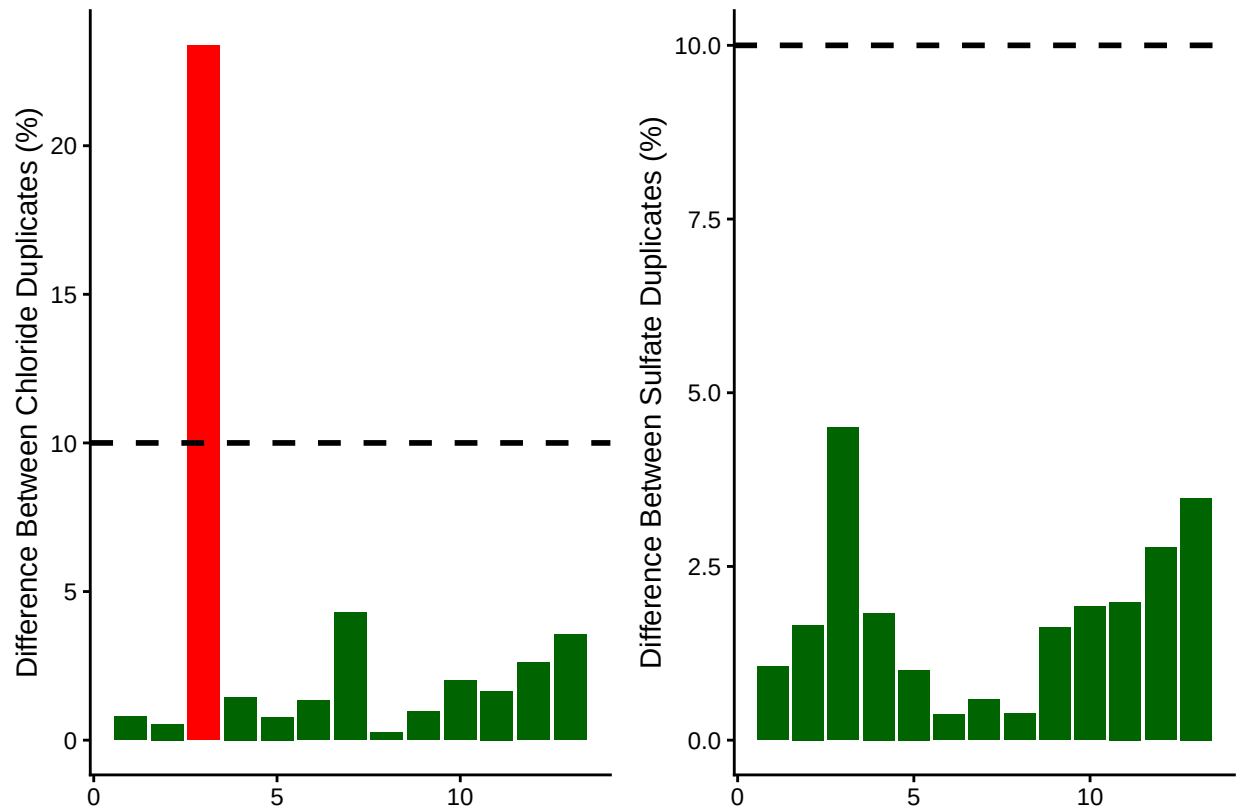
```
## [1] 0.002142857
```

## 0.5 Assess Duplicates



```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

## 0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$S04_Conc_mM <- (all_dat$S04_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for TEMPEST this depends on the sample so...
all_dat$Dilution <- 50
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "TMP"), 50, all_dat$Dilution)
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "EST_SourceWater"), 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "SW_SourceWater"), 100, all_dat$Dilution)
```

```

all_dat$Dilution <- ifelse(str_detect(all_dat$sample_ID, "FATE"), 100, all_dat$Dilution)

all_dat$Dilution <- ifelse(all_dat$sample_ID == "90", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "92", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "93", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "94", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "129", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "132", 100, all_dat$Dilution)

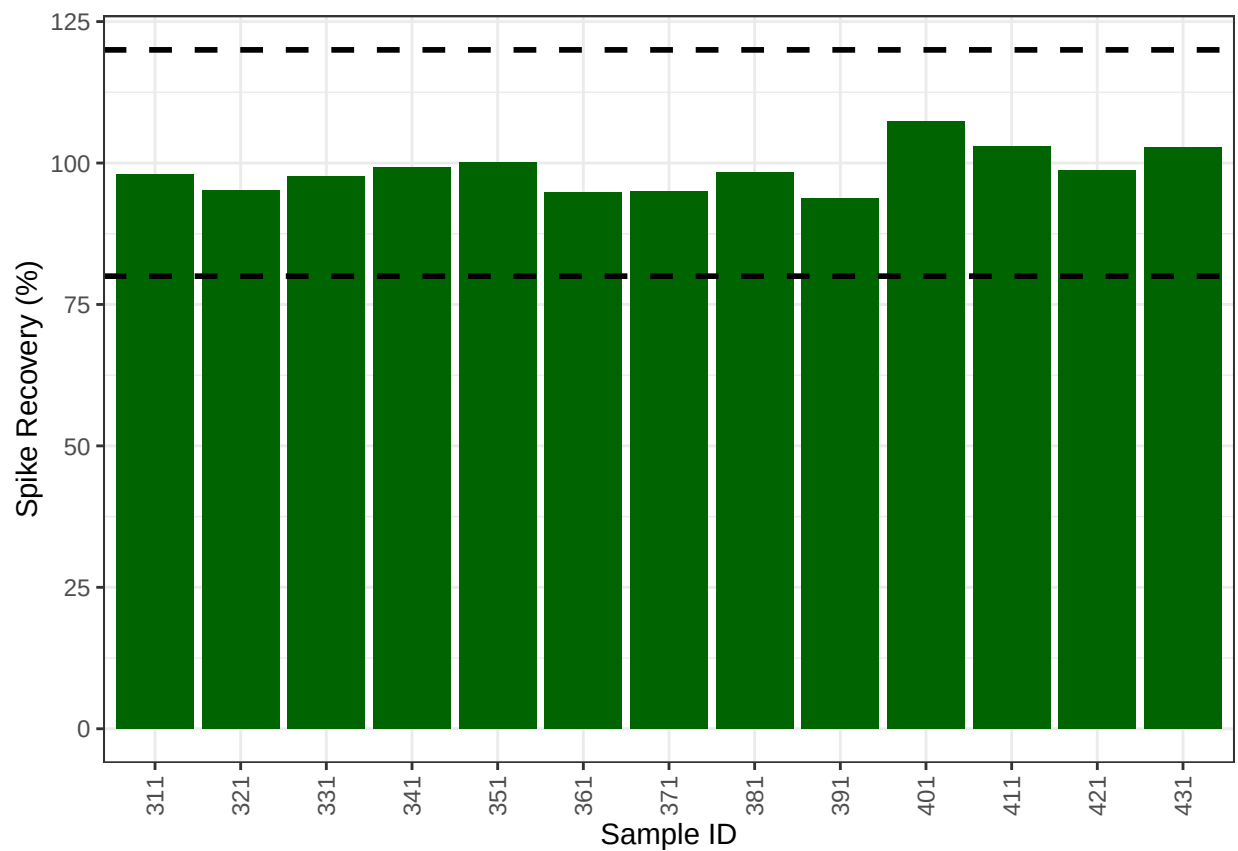
all_dat$Dilution <- ifelse(all_dat$sample_ID == "169", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "171", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "172", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "205", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "207", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "208", 100, all_dat$Dilution)

all_dat$Dilution <- ifelse(all_dat$sample_ID == "294", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "296", 100, all_dat$Dilution)
all_dat$Dilution <- ifelse(all_dat$sample_ID == "297", 100, all_dat$Dilution)

# head(all_dat)

```

## 0.7 Assess Analytical Spikes



```
## [1] ">80% of SO4 spikes have a recovery between the high and low cutoff - PROCEED"
```

## 0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```

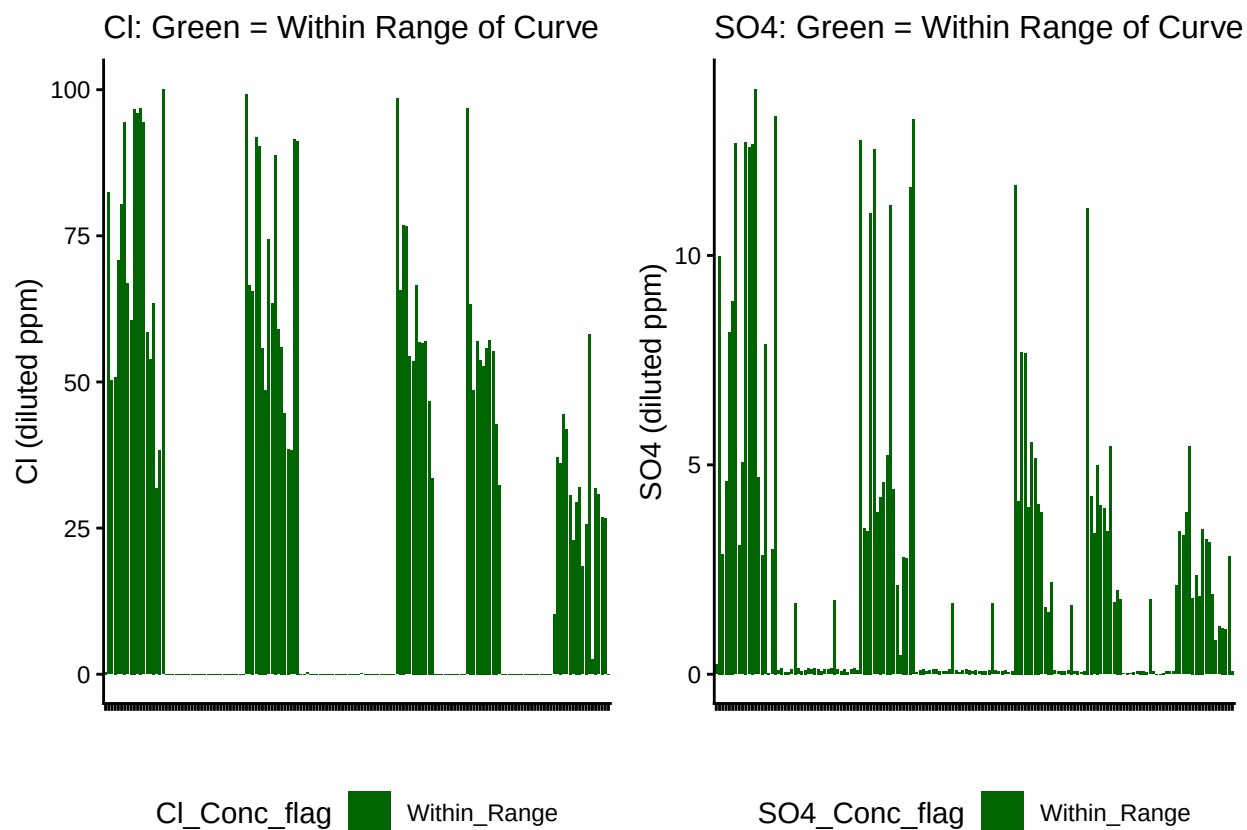


Table 1: SO4 samples

| SO4_Conc_flag | Percent_samples |
|---------------|-----------------|
| Within_Range  | 100             |

Table 2: Cl samples

| Cl_Conc_flag | Percent_samples |
|--------------|-----------------|
| Within_Range | 100             |

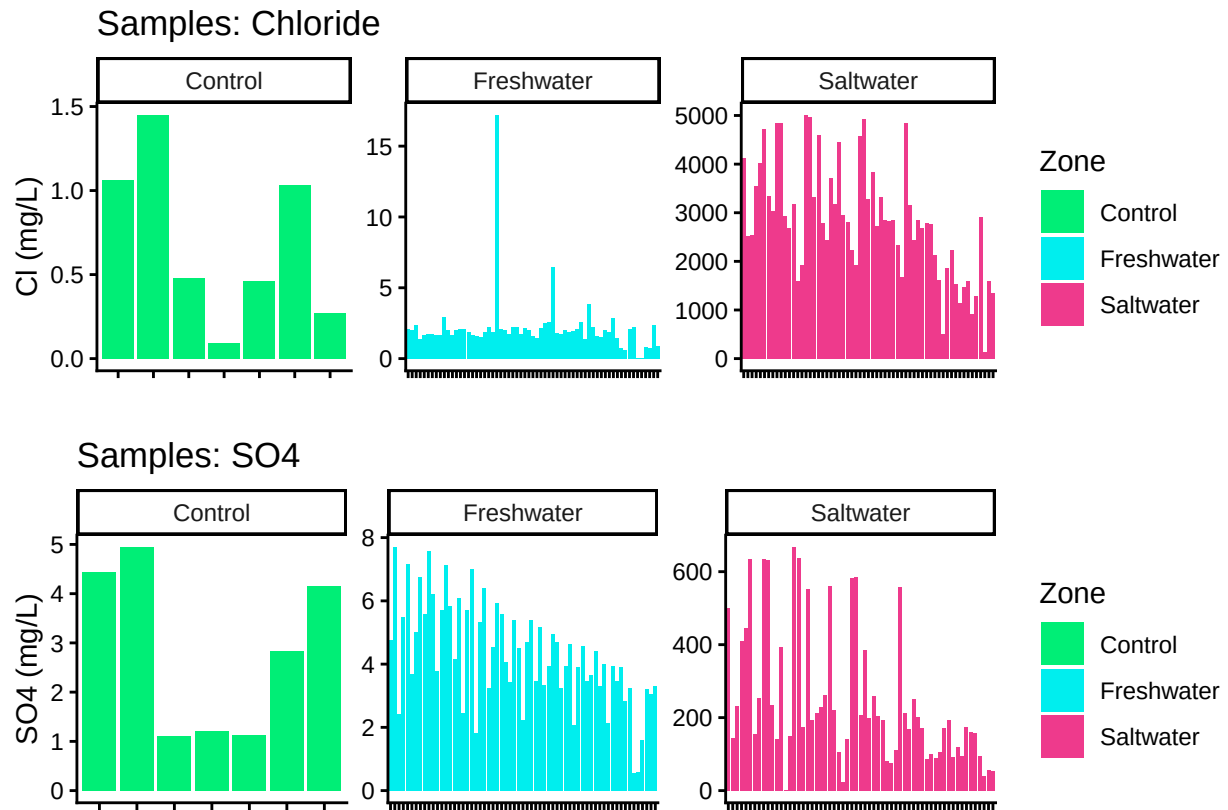
```
##Merge with Sample ID list to get actual sample IDs
```

```
##Pull in active porewater inventory
```

```
##Check samples against metadata
```

```
## ***All sample IDs are present in metadata.***
```

## 0.9 Visualize Data by Plot



## 0.10 Export Processed Data

#end